

Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)



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Abstract

Falls, frailty, and chronic conditions such as cardiovascular disease, diabetes, and obesity remain leading causes of morbidity and mortality in older adults. Yet, existing predictive models often rely narrowly on physical health indicators, overlooking nutrition and lifestyle factors that may enhance early detection of risk. This thesis develops and evaluates machine learning models using data from The Irish Longitudinal Study on Ageing (TILDA) to predict key health outcomes in a cohort of older adults. The models integrate physical, mental, lifestyle, and socio-demographic predictors, with particular emphasis on under-explored nutritional and behavioural variables highlighted in prior research. Multiple approaches are compared, including logistic regression, random forests, and gradient boosting, with performance assessed via cross-validation and feature importance analysis. By identifying high-impact, modifiable predictors, this work aims to advance early risk stratification, guide targeted prevention strategies, and inform public health policy to support healthy ageing.

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

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Limerick, 2026

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Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

AI	Artificial Intelligence. 1
AUC	Area Under the ROC Curve. 4, 9
DNN	Deep Neural Network. 8
GBM	Gradient Boosting Machine. 3, 8
HRS	Health and Retirement Study, (a longitudinal study of ageing in the United States). 2
LDA	Linear Discriminant Analysis. 8
ML	Machine Learning. 1, 3, 4
MPOD	Macular pigment optical density. 8
RF	Random Forest. 3, 8
SHAP	Shapley Additive Explanations. 9
TILDA	The Irish Longitudinal Study on Ageing. 2–5
WHO	World Health Organization. 7

List of Abbreviations

Chapter 1

Introduction

1.1 Background and Context

Chronic conditions—including cardiovascular disease, diabetes, and frailty—are leading contributors to morbidity and mortality in ageing populations [1, 2]. These conditions are associated with declines in functional capacity (e.g., gait speed, grip strength [3]), increased healthcare utilisation [4], and reduced quality of life [5].

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non-communicable diseases, contributing billions in healthcare costs annually [6]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [7]. Recognising this, international and national policies—including the World Health Organization’s Global Action Plan on Noncommunicable Diseases and Ireland’s *Healthy Ireland* framework—highlight nutrition and lifestyle interventions as central to healthcare strategy.

In parallel, artificial intelligence (Artificial Intelligence (AI)) and machine learning (Machine Learning (ML)) are increasingly applied in healthcare to support risk prediction and personalised intervention. However, European regulations such as the General Data Protection Regulation (GDPR) and the forthcoming EU Artificial Intelligence Act impose strict requirements for transparency, explainability, and governance. This regulatory context underscores the importance of interpretable and ethically robust ML approaches.

1. INTRODUCTION

Traditional risk prediction models have largely relied on clinical biomarkers such as blood pressure and lipid profiles. Yet, growing evidence shows that broader predictors—including nutrition [8], physical activity [9], mental health [10], and social determinants [11]—enhance predictive performance.

The Irish Longitudinal Study on Ageing (The Irish Longitudinal Study on Ageing (TILDA)) offers a nationally representative dataset of adults aged 50+ in Ireland with repeated measures across multiple domains. The Harmonized TILDA dataset aligns variables with the RAND Health and Retirement Study (Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS)) framework to ensure cross-country comparability, while raw TILDA waves retain study-specific variables. The IDS dataset further extends coverage to participants with disabilities. Together, these resources provide both consistency for international comparison and flexibility to capture detailed, study-specific measures.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over-reliance on clinical predictors, limited incorporation of lifestyle and social variables, predominantly cross-sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by integrating TILDA’s longitudinal, multidomain data with interpretable ML methods, demonstrating how nutrition, physical activity, and mental health can be combined with traditional clinical markers to improve prediction of ageing-related outcomes.

1.2 Research Problem and Motivation

Despite increasing policy interest in lifestyle and nutrition and advances in ML methods, approaches to chronic disease prediction in older adults remain constrained in several ways:

1.2.1 Limited Integration of Multidimensional Data

Most models prioritise clinical variables while underutilising modifiable lifestyle and nutrition factors [8, 12].

1.2.2 Underuse of Longitudinal Dynamics

Few studies exploit repeated measures to capture trajectories of risk, such as frailty progression [13].

1.2.3 Lack of Explainability

Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthy AI in healthcare [14].

Addressing these challenges is essential to improve predictive accuracy, support evidence-based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.3 Research Aims and Objectives

This thesis develops and validates ML models that integrate nutrition, lifestyle, and mental health features with clinical data to improve prediction of chronic disease and ageing-related outcomes in a representative cohort of older adults.

- Review existing applications of ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.
- Preprocess and harmonise TILDA waves 1-5, and engineer multidomain features (nutrition, activity, mobility, mental health, clinical indicators).
- Develop and compare baseline statistical models (e.g., logistic regression) with advanced ML methods (Random Forest (RF), Gradient Boosting Machine (GBM)).

1. INTRODUCTION

- Evaluate models using discrimination (Area Under the ROC Curve (AUC)), calibration, and error metrics (accuracy, precision, recall, F1-score); perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.
- Identify actionable predictors (e.g., vitamin D status, gait speed reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.4 Contributions of the Research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how nutrition, lifestyle, and mental health features enhance prediction when combined with established clinical baselines.
- **Methodological:** Provides a reproducible pipeline for harmonising longitudinal TILDA data and integrating multidomain features into ML workflows.
- **Clinical:** Identifies modifiable predictors with relevance for targeted interventions and policy to promote healthy ageing.

1.5 Thesis Outline

- *Chapter Two:* Literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle and nutrition predictors and methodological considerations.
- *Chapter Three:* TILDA study design, data preprocessing, harmonisation, and feature engineering.
- *Chapter Four:* Modelling framework, including baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics.

- *Chapter Five*: Results, model comparison, feature importance, and ablation analyses.
- *Chapter Six*: Discussion of implications, limitations, ethical considerations, and future directions.

Supporting documents are included in the appendix:

- *Appendix A*: Variable codebook and wave harmonisation notes (including anonymisation actions, top-coding, and grouping rules relevant to the features used).
- *Appendix B*: Ethics.
- *Appendix C*: Modelling details, supplementary tables/figures, and reproducibility checklist (overview of data processing scripts).

A digital archive accompanies this dissertation (access request required):

- *Folder 1*: Preprocessing and feature-engineering scripts (documentation only; no raw TILDA data).
- *Folder 2*: Model training/evaluation code and configuration files to reproduce reported results.

1. INTRODUCTION

Chapter 2

Literature Review

2.1 Introduction

- Purpose of this chapter
- Brief overview on TILDA as a unique dataset (rich, longitudinal, multi-domai...)
- Scope: frailty, mental health, chronic disease, multimorbidity, nutrition, and ML models

2.2 Chronic Disease and Ageing

- The burden of chronic conditions in older adults
- Healthy ageing framework World Health Organization (WHO)
- Importance of multimorbidity, frailty, mental health

2.2.1 Frailty and Physical Function

- Gait speed, grip strength, functional mobility decline
- Predictors of frailty transitions ([15], [16], [17])

2. LITERATURE REVIEW

2.2.2 Mental Health in Older Adults

- Depression, anxiety, loneliness, social connectedness ([18], [19], [20])
- Links between physical activity and depression ([10], [9], [12])

2.2.3 Nutrition and Biological Ageing

- Vitamin D, lutein, zeaxanthin, Macular pigment optical density (MPOD) ([8], [21], lairdVitaminDeficiencyIreland2020, feeneyLowMacularPigment2013)
- Malnutrition and its predictors ([2])
- Biological vs chronological ageing ([22], [23])

2.3 Machine Learning in Ageing and Chronic Disease

- Overview: Why ML is valuable in ageing research [24]
- Transition from regression-based models → ensemble ML → deep learning(cite a example here)

2.3.1 Feature Sets in ML Studies

- Clinical, sociodemographic, lifestyle, cognition, psychosocial
- Over-reliance on physical health predictors vs underuse of nutrition/social variables

2.3.2 ML Models Used

- Logistic regression, Markov models, Linear Discriminant Analysis (LDA) (traditional)
- Ensembles RF, GBM, Deep Neural Network (DNN) – e.g., ([25]) diabetes prediction

- Unsupervised clustering for dementia ([26])
- Brain age prediction models ([27], [28])

2.3.3 Performance and Validation

- AUC, R^2 ranges across tasks (diabetes, gait speed, mortality, depression)
- Limitations: internal validation only, cross-sectional data, no generalisation

2.3.4 Explainability in ML for Ageing

- Shapley Additive Explanations (SHAP), TreeExplainer examples ([25], [29], [3])
- Challenges of black-box models in clinical practice

2.4 Gaps in the Literature

- Limited population diversity (mostly Irish, 50+)
- Over-reliance on physical predictors; neglect of nutrition, social, religious/spiritual, environmental variables
- Dominance of regression models; limited advanced ML such as neural networks, transformers
- Validation gaps – very few external validations, especially for ensemble and brain-age models
- Underexplored outcomes: pain, sleep, QoL
- Cross-sectional bias; weak time-series modelling
- Poor transparency on missing data handling
- Limited integration of nutrition biomarkers
- Lack of biological age prediction vs chronological

2. LITERATURE REVIEW

- Interpretability challenges in ML

2.5 Chapter Summary

- I need to recap the main findings from LR systematic mapping excel file
- Identify your contribution niche:
 - integrating nutrition + lifestyle + clinical predictors
 - ML + explainability
 - time-based prediction (longitudinal TILDA data)
- Lead into Methodology (Chapter 3)

Chapter 3

Data and Preprocessing

3.1 Insert a Figure

We refer to a figure in the text like this: (Figure 3.1).



Figure 3.1: Title of Figure - Description. [Source: (?)]

3. DATA AND PREPROCESSING

3.2 Creating a list

- (2000) item 1.
- (2004) item 2.
- (2010) item 3.
- (2013) item 4.

3.2.1 Creating a Table

Table 3.1: Table Title

	Resolution	Min	Max
Gyroscope	4.5mV / °/s	0.27 °/s	406 °/s
Accelerometer	600mV /g	0.002g	2g
Magnetometer	385mV/ gauss	0.317 gauss	6 gauss

3.3 Quote

An in-text quote is declared in the following way:

Recycling is the way forward! Recycling is the way forward! Recycling
is the way forward! Recycling is the way forward! Recycling is the
way forward! Recycling is the way forward! Recycling is the way
forward! Recycling is the way forward! Recycling is the way forward!
Recycling is the way forward! Recycling is the way forward! Recycling
is the way forward! Recycling is the way forward! Recycling is the
way forward! Recycling is the way forward! Recycling is the way
forward! Recycling is the way forward! Recycling is the way forward!
Recycling is the way forward! (CommonSense, p. 0)

Another way of doing it is:

*There are in our existence spots of time,
That with distinct pre-eminence retain
A renovating virtue, whence-depressed
By false opinion and contentious thought,
Or aught of heavier or more deadly weight,
In trivial occupations, and the round
Of ordinary intercourse-our minds
Are nourished and invisibly repaired;
A virtue, by which pleasure is enhanced,
That penetrates, enables us to mount,
When high, more high, and lifts us up when fallen.*

(? , verses 208-218)

3. DATA AND PREPROCESSING

Chapter 4

Methodology

4.1 Math

In-text Math

- A vector vector $\hat{A}_{ba}\{x_{ba}, y_{ba}, z_{ba}\}$
- Another vector $R^b\{\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}\}$ as:

A series of Equations:

$$\hat{t}_{1b} = \hat{A}_{ba} \tag{4.1}$$

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{|\hat{A}_{ba} \times \hat{M}_{bm}|} \tag{4.2}$$

$$\hat{t}_{3b} = \hat{t}_{1b} \times \hat{t}_{2b} \tag{4.3}$$

$$R^a = R^b(R^i)^T = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}][\hat{t}_{1i}, \hat{t}_{2i}, \hat{t}_{3i}]^T \tag{4.4}$$

where the symbol T denotes transposition.

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{|\hat{A}_{ba} \times \hat{M}_{bm}|} \tag{4.5}$$

4. METHODOLOGY

$$\begin{aligned}
&= \frac{(y_{ba}z_{bm} - y_{bm}z_{ba}, z_{ba}x_{bm} - z_{bm}x_{ba}, x_{ba}y_{bm} - x_{bm}y_{ba})}{|A_{ba}| |M_{bm}| \sqrt{1 - (\hat{A}_{ba} \cdot \hat{M}_{bm})^2}} \\
&= \frac{-0.2338, -0.0931, -0.0651}{0.2601} \\
&= (-0.8989, -0.3579, -0.2503)
\end{aligned}$$

which if normalised gives:

$$= (-0.8995, -0.3581, -0.2504)$$

A matrix

$$R^b = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}] = \begin{pmatrix} -0.2698 & -0.8995 & 0.3439 \\ 0.0035 & -0.3581 & -0.9337 \\ 0.9629 & -0.2504 & 0.0997 \end{pmatrix}$$

Chapter 5

Results

5.1 References

Open the file ‘references_myphd’ with BibTex. The file is located in ‘/7_references/’

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book:** (? , p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter:** (?)
- **Book Contribution:** (?)
- **MUSIC CD/DVD:** (?)
- **Journal Article:** (?)
- **Patent:** (?)
- **PhD/Master Dissertation:** (?)
- **Conference Paper:** (?)
- **Technical Report:** (?)

5. RESULTS

- **Unpublished Report:** (?)
- **Web Videos**(?) (. . . not the name of the person who uploaded the video though)
- **Web Paper:** (?)
- **Web Journal Article:** (?)
- **Webpage:** (?)
- **Downloaded Images or Sounds:** (?)

You can also use this kind of referencing if needed in the text:
Surname [?] said that . . .

Chapter 6

Conclusions and Future Directions

6.1 Summary

6.2 Conclusions

6.3 Contributions

6.4 Future Work

6. CONCLUSIONS AND FUTURE DIRECTIONS

Appendix A

Insert a figure

Appendix

Appendix B

Title

Type here you text as usual . . .

Appendix

Appendix C

Code for estimating attitude

AHRS_TRIAD.C

```
/*
 *
 * Copyright (c) 2013, Giuseppe Torre
 * All rights reserved.
 *
 * Redistribution and use in source and binary forms, with or without
 * modification, are permitted provided that the following conditions are met:
 *     * Redistributions of source code must retain the above copyright
 *       notice, this list of conditions and the following disclaimer.
 *     * Redistributions in binary form must reproduce the above copyright
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 * (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES;
 * LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND
 * ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
 * (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
 * SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 */

#include "ext.h"
#include "ext.mess.h"
#include <string.h>
#include <stdio.h>
#include <math.h>
#define EPSILON 1e-6

//*****CALIBRATION STUFF*****//
/* INTRODUCE THE OFFSET VALUES IN THE FOLLOWING VARIABLES IN ADC values */
#define ACCELX.OFFSET 2043 //initial offset in accelerometer X
#define ACCELY.OFFSET 2053 // initial offset in accelerometer Y
#define ACCELZ.OFFSET 2264//initial offset in accelerometer Z

#define MagX.OFFSET 1917 //initial offset in Magnetometer X
```

Appendix

```
#define MagY_OFFSET 1813 // initial offset in Magnetometer Y
#define MagZ_OFFSET 1667 // initial offset in Magnetometer Z
/* WE SHOULD CALIBRATE THE FOLLOWING VALUES FOR EACH PARTICULAR IMU axis */
#define ACCELX_Resolution 536
#define ACCELY_Resolution 661
#define ACCELZ_Resolution 559
#define MagX_Resolution 186
#define MagY_Resolution 189
#define MagZ_Resolution 170
// ***** END CALIBRATION STUFF *****//

void *this_class; // Required. Global pointing to this class

typedef struct _triad // Data structure for this object
{
    t_object m_ob; // Must always be the first field; used by Max

    Atom m_args[9]; // we want our inlet to be receiving a list of 10 elements

    long m_value; //inlet

    void *m_R1; //these are all the outlets for the 3 X 3 Matrix
    void *m_R2; // R1 -> firts top left cell ..R2 middle cell of the first
    row ...and so on
    void *m_R3; //
    void *m_R4; // End Outlets
} t_triad;

void *triad_new(long value);

void triad_assist(t_triad *triad, void *b, long msg, long arg, char *
s);
void triad_free(t_triad *triad);

void triad_list(t_triad *x, Symbol *s, short argc, t_atom *argv);

void MatrixByMatrix(double *Result, double *MatrixLeft, double *
MatrixRight);

void Matrix2Quat(double *Quat, double *Matrix);
void Quat2Matrix(double *Matrix, double *Quat);
void inverseQuat(double *InvQuat, double *RegQuat);
void NormQuat(double *YesQuat, double *NotQuat);
void Slerp(double *NewQuat, double *OldQuat, double *CurrentQuat);
void NormVect(double *YesVect, double *NotVect);
double orientationMatrix[9];
double Result[9];
int i;
double InvorientationMatrix[9];

double temp[6];
double ref[6];
double vectAx[3];
double vectAy[3];
double vectAz[3];
double vectBx[3];
double vectBy[3];
double vectBz[3];
double MagnCrosProd_A;
double MagnCrosProd_B;
double accnorm, magnorm, earthnorm, VectAynorm, VectAznorm,
VectBynorm, VectBznorm;
```

```

        double m[9], n[9];
        double quat_e[4];
        double invquat_e[4];
        double mult;
        double quat_new[4];
        double quat_old[4];
        double ecs, accex_ADCnumber, y, accey_ADCnumber, z, accez_ADCnumber, mx,
            magnx_ADCnumber, my, magny_ADCnumber, mz, magnz_ADCnumber;

// SLERP Variables

        double trace, Suca;
        double tol[4], omega, sinom, cosom, scale0, scale1, tez,
            orientationMatrixA[9];

int main(void)
{
    // set up our class: create a class definition
    setup((t_messlist**) &this_class, (method)triad_new, (method)triad_free, (short)
        sizeof(t_triad), 0L, A_GIMME, 0);
    address((method)triad_list, "list", A_GIMME, 0);
    address((method)triad_assist, "assist", A_CANT, 0);
    finder_addclass("Maths", "triad");
    post(".... I 'm-TRIAD-Object !.... from -AHRS- Library ...", 0);
    return 0;
}

/* ----- triad_new ----- */

void *triad_new(long value)
{
    t_triad *triad;
    triad = (t_triad *)newobject(this_class);           // create the new instance and return
        a pointer to it
    triad->m_R4 = floatout(triad);
    triad->m_R3 = floatout(triad);
    triad->m_R2 = floatout(triad);
    triad->m_R1 = floatout(triad);

    return(triad);
}

etc.

etc.

```

Appendix

References

- [1] Leonardo Bencivenga, Klara Komici, Pierangela Nocella, Fabrizio Vincenzo Grieco, Angela Spezzano, Brunella Puzone, Alessandro Cannavo, Antonio Cittadini, Graziamaria Corbi, Nicola Ferrara, and Giuseppe Rengo. Atrial fibrillation in the elderly: A risk factor beyond stroke. *Ageing Research Reviews*, 61:101092, 2020. ISSN 1568-1637. doi: 10.1016/j.arr.2020.101092. 1
- [2] Laura A Bardon, Melanie Streicher, Clare A Corish, Michelle Clarke, Lauren C Power, Rose Anne Kenny, Deirdre M O'Connor, Eamon Laird, Eibhlis M O'Connor, Marjolein Visser, Dorothee Volkert, Eileen R Gibney, and MaNuEL Consortium. Predictors of Incident Malnutrition in Older Irish Adults from the Irish Longitudinal Study on Ageing Cohort—A MaNuEL study. *J Gerontol A Biol Sci Med Sci*, 75(2):249–256, January 2020. ISSN 1079-5006. doi: 10.1093/gerona/gly225. 1, 8
- [3] James R. C. Davis, Silvin P. Knight, Orna A. Donoghue, Belinda Hernández, Rossella Rizzo, Rose Anne Kenny, and Roman Romero-Ortuno. Comparison of Gait Speed Reserve, Usual Gait Speed, and Maximum Gait Speed of Adults Aged 50+ in Ireland Using Explainable Machine Learning. *Front Netw Physiol*, 1:754477, 2021. ISSN 2674-0109. doi: 10.3389/fnetp.2021.754477. 1, 9
- [4] Patrick V. Moore, Kathleen Bennett, and Charles Normand. Counting the time lived, the time left or illness? Age, proximity to death, morbidity and prescribing expenditures. *Social Science & Medicine*, 184:1–14, 2017. ISSN 0277-9536. doi: 10.1016/j.socscimed.2017.04.038. 1
- [5] Ying Huang, Yuhao Su, Ying Jiang, and Meilan Zhu. Sex differences in the associations between blood pressure and anxiety and depression scores in a middle-aged and elderly population: The Irish Longitudinal Study on Ageing (TILDA). *Jour-*

REFERENCES

- nal of Affective Disorders*, 274:118–125, September 2020. ISSN 0165-0327. doi: 10.1016/j.jad.2020.05.133. 1
- [6] World Health Organization Europe. New data: Noncommunicable diseases cause 1.8 million avoidable deaths and cost US\$ 514 billion every year, reveals new WHO/Europe report. <https://www.who.int/europe/news/item/27-06-2025-new-data-noncommunicable-diseases-cause-1-8-million-avoidable-deaths-and-cost-us-514-billion-USD-every-year-reveals-new-who-europe-report>, 2025. 1
- [7] Niamh Rice and Charles Normand. The cost associated with disease-related malnutrition in Ireland. *Public Health Nutr*, 15(10):1966–1972, October 2012. ISSN 1475-2727. doi: 10.1017/S1368980011003624. 1
- [8] Caoileann H. Murphy, Eoin Duggan, James Davis, Aisling M. O’Halloran, Silvin P. Knight, Rose Anne Kenny, Sinead N. McCarthy, and Roman Romero-Ortuno. Plasma lutein and zeaxanthin concentrations associated with musculoskeletal health and incident frailty in The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 171:112013, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2022.112013. 2, 3, 8
- [9] Eamon Laird, Charlotte Lund Rasmussen, Rose Anne Kenny, and Matthew P. Herring. Physical Activity Dose and Depression in a Cohort of Older Adults in The Irish Longitudinal Study on Ageing. *JAMA Netw Open*, 6(7):e2322489, July 2023. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2023.22489. 2, 8
- [10] C. P. McDowell, R. K. Dishman, M. Hallgren, C. MacDonncha, and M. P. Herring. Associations of physical activity and depression: Results from the Irish Longitudinal Study on Ageing. *Experimental Gerontology*, 112:68–75, 2018. ISSN 0531-5565. doi: 10.1016/j.exger.2018.09.004. 2, 8
- [11] Cathal McCrory, Ciaran Finucane, Celia O’Hare, John Frewen, Hugh Nolan, Richard Layte, Patricia M. Kearney, and Rose Anne Kenny. Social Disadvantage and Social Isolation Are Associated With a Higher Resting Heart Rate: Evidence From The Irish Longitudinal Study on Ageing. *J Gerontol B Psychol Sci Soc Sci*, 71(3):463–473, May 2016. ISSN 1079-5014. doi: 10.1093/geronb/gbu163. 2
- [12] Eamon Laird, Matthew P. Herring, Brian P. Carson, Catherine B. Woods, Cathal Walsh, Rose Anne Kenny, and Charlotte Lund Rasmussen. Physical activity for

REFERENCES

- depression among the chronically ill: Results from older diabetics in the Irish longitudinal study on ageing. *Psychiatry Research*, 326:115274, August 2023. ISSN 0165-1781. doi: 10.1016/j.psychres.2023.115274. 3, 8
- [13] Sebastian Moguilner, Silvin P. Knight, James R. C. Davis, Aisling M. O’Halloran, Rose Anne Kenny, and Roman Romero-Ortuno. The Importance of Age in the Prediction of Mortality by a Frailty Index: A Machine Learning Approach in the Irish Longitudinal Study on Ageing. *Geriatrics (Basel)*, 6(3):84, August 2021. ISSN 2308-3417. doi: 10.3390/geriatrics6030084. 3
- [14] Orna A Donoghue, Belinda Hernandez, Matthew D L O’Connell, and Rose Anne Kenny. Using Conditional Inference Forests to Examine Predictive Ability for Future Falls and Syncope in Older Adults: Results from The Irish Longitudinal Study on Ageing. *J Gerontol A Biol Sci Med Sci*, 78(4):673–682, April 2023. ISSN 1758-535X. doi: 10.1093/gerona/glac156. 3
- [15] Kevin McCarthy, Eamon Laird, Aisling M. O’Halloran, Padraic Fallon, Román Romero Ortuño, and Rose Anne Kenny. Association between metabolic syndrome and risk of both prevalent and incident frailty in older adults: Findings from The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 172:112056, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2022.112056. 7
- [16] Chuanying Huang, Shuqin Sun, Xiaocao Tian, Tong Wang, Tianyu Wang, Haiping Duan, and Yili Wu. Age modify the associations of obesity, physical activity, vision and grip strength with functional mobility in Irish aged 50 and older. *Archives of Gerontology and Geriatrics*, 84:103895, 2019. ISSN 0167-4943. doi: 10.1016/j.archger.2019.05.020. 7
- [17] Matthew D. L. O’Connell, George M. Savva, Chie Wei Fan, and Rose Anne Kenny. Orthostatic hypotension, orthostatic intolerance and frailty: The Irish Longitudinal Study on Aging-TILDA. *Archives of Gerontology and Geriatrics*, 60(3):507–513, 2015. ISSN 0167-4943. doi: 10.1016/j.archger.2015.01.008. 7
- [18] Ziggi Ivan Santini, Ai Koyanagi, Stefanos Tyrovolas, Josep Maria Haro, Robert J. Donovan, Line Nielsen, and Vibeke Koushede. The protective properties of Act-Belong-Commit indicators against incident depression, anxiety, and cognitive impairment among older Irish adults: Findings from a prospective community-based study. *Experimental Gerontology*, 91:79–87, 2017. ISSN 0531-5565. doi: 10.1016/j.exger.2017.02.074. 8

REFERENCES

- [19] Celeste de Jager and Ann Marie Haigh. Symposia. *International Psychogeriatrics*, 23:S12–S96, 2011. ISSN 1041-6102. doi: 10.1017/S104161021100127X. 8
- [20] Robert Briggs, Mark Ward, and Rose Anne Kenny. The ‘Wish to Die’ in later life: Prevalence, longitudinal course and mortality. Data from TILDA. *Age and Ageing*, 50(4):1321–1328, July 2021. ISSN 0002-0729. doi: 10.1093/ageing/afab010. 8
- [21] Kevin McCarthy, Eamon Laird, Aisling M. O’Halloran, Cathal Walsh, Martin Healy, Annette L. Fitzpatrick, James B. Walsh, Belinda Hernández, Padraic Fallon, Anne M. Molloy, and Rose Anne Kenny. Association between vitamin D deficiency and the risk of prevalent type 2 diabetes and incident prediabetes: A prospective cohort study using data from The Irish Longitudinal Study on Ageing (TILDA). *EClinicalMedicine*, 53:101654, November 2022. ISSN 2589-5370. doi: 10.1016/j.eclinm.2022.101654. 8
- [22] Kevin McCarthy, Aisling M. O’Halloran, Padraic Fallon, Rose Anne Kenny, and Cathal McCrory. Metabolic syndrome accelerates epigenetic ageing in older adults: Findings from The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 183:112314, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2023.112314. 8
- [23] Zahra Azizi, Rebecca J. Hirst, Alan O’ Dowd, Cathal McCrory, Rose Anne Kenny, Fiona N. Newell, and Annalisa Setti. Evidence for an association between allostatic load and multisensory integration in middle-aged and older adults. *Archives of Gerontology and Geriatrics*, 116:105155, 2024. ISSN 0167-4943. doi: 10.1016/j.archger.2023.105155. 8
- [24] Zuo Zhang, Lauren Robinson, Robert Whelan, Lee Jollans, Zijian Wang, Frauke Nees, Congying Chu, Marina Bobou, Dongping Du, Ilinca Cristea, Tobias Banaschewski, Gareth J. Barker, Arun L. W. Bokde, Antoine Grigis, Hugh Garavan, Andreas Heinz, Rüdiger Brühl, Jean-Luc Martinot, Marie-Laure Paillère Martinot, Eric Artiges, Dimitri Papadopoulos Orfanos, Luise Poustka, Sarah Hohmann, Sabina Millenet, Juliane H. Fröhner, Michael N. Smolka, Nilakshi Vaidya, Henrik Walter, Jeanne Winterer, M. John Broulidakis, Betteke Maria van Noort, Argyris Stringaris, Jani Penttilä, Yvonne Grimmer, Corinna Insensee, Andreas Becker, Yuning Zhang, Sinead King, Julia Sinclair, Gunter Schumann, Ulrike Schmidt, and Sylvane Desrivieres. Machine learning models for diagnosis and risk prediction

REFERENCES

- in eating disorders, depression, and alcohol use disorder. *Journal of Affective Disorders*, 379:889–899, June 2025. ISSN 0165-0327. doi: 10.1016/j.jad.2024.12.053. 8
- [25] Xuezhong Xu, Xue Mingyang, Jie Yang, Hailong Zheng, and Zhifei Che. Tailored machine learning for evaluating the long-term diabetes risk in older individuals: Findings from the Irish Longitudinal Study on Ageing (TILDA). *BMJ Open*, 13(5):e072991, May 2023. ISSN 2044-6055. doi: 10.1136/bmjopen-2023-072991. 8, 9
- [26] Bayen, Eleonore, Laurent, Cleret de Langavan, and Yaffe, Kristine. Unsupervised Machine Learning to Identify High Likelihood of Dementia in Population-Based Surveys: Development and Validation Study. *Journal of Medical Internet Research*, 20(7), July 2018. ISSN 1438-8871. doi: 10.2196/10493. 9
- [27] Rory Boyle, Lee Jollans, Laura M. Rueda-Delgado, Rossella Rizzo, Görsev G. Yener, Jason P. McMorrow, Silvin P. Knight, Daniel Carey, Ian H. Robertson, Derya D. Emek-Savaş, Yaakov Stern, Rose Anne Kenny, and Robert Whelan. Brain-predicted age difference score is related to specific cognitive functions: A multi-site replication analysis. *Brain Imaging Behav*, 15(1):327–345, February 2021. ISSN 1931-7565. doi: 10.1007/s11682-020-00260-3. 9
- [28] Morgana A. Shirsath, John D. O’Connor, Rory Boyle, Louise Newman, Silvin P. Knight, Belinda Hernandez, Robert Whelan, James F. Meaney, and Rose Anne Kenny. Slower speed of blood pressure recovery after standing is associated with accelerated brain ageing: Evidence from The Irish Longitudinal Study on Ageing (TILDA). *Cerebral Circulation - Cognition and Behavior*, 6:100212, January 2024. ISSN 2666-2450. doi: 10.1016/j.cccb.2024.100212. 9
- [29] James Davis, Silvin P. Knight, Rossella Rizzo, Orna A. Donoghue, Rose Anne Kenny, and Roman Romero-Ortuno. A linear regression-based machine learning pipeline for the discovery of clinically relevant correlates of gait speed reserve from multiple physiological systems. In *2021 29th European Signal Processing Conference (EUSIPCO)*, pages 1266–1270, August 2021. doi: 10.23919/EUSIPCO54536. 2021.9616187. 9